

SUDIP BASU

Senior Python Developer & Automation Engineer | Cloud (AWS), Flutter & Rust

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 github.com/piidus

 [LinkedIn](#)

 [Replit](#)

Work Availability: Available **strictly for 100% Remote positions or Freelancing / Contractual engagements** due to a bedridden medical condition. Equipped with a high-performance workstation, fiber internet, power backup, and immediate full-time availability.

PROFESSIONAL SUMMARY

Results-oriented **Python Developer, Automation Specialist & Multi-Stack Engineer** with **6+ years of hands-on experience** architecting resilient web automation pipelines, backend RESTful services, cloud deployments on AWS, and cross-platform applications. Strong polyglot capabilities spanning **Python, Rust** (systems concurrency & performance), and **Flutter / Dart** (cross-platform mobile applications). Proven track record building algorithmic trading engines (ICICI Securities Breeze API), stateful RPA bots, and open-source libraries. Early adopter of **AI-assisted software engineering** using **Google Gemini** and **Antigravity IDE** to accelerate robust, error-handled code delivery.

TECHNICAL SKILLS

Languages & Core	Python 3 (Advanced), Rust, Dart, SQL, Bash / PowerShell, HTML5, CSS3, JavaScript
Mobile & Cross-Platform	Flutter (Widgets, Routing, Cross-Platform UI, State Management), Flet, BeeWare, Kivy
Systems & Concurrency	Rust (Cargo, Memory Safety, Ownership/Borrowing, Concurrency), Python Multi-threading
Automation & Scraping	Selenium WebDriver, Playwright, BeautifulSoup4, Requests, Headless Browsers, Session Resumption, Anti-Bot Handling
Cloud & DevOps (AWS)	AWS (EC2, S3, IAM, CloudWatch, Lambda), cPanel, Ubuntu/Linux Server Admin, Git, GitHub Actions
Web & APIs	Flask, REST API Design & Integration, Role-Based Access Control (RBAC), SQLite, MySQL, PostgreSQL
Data & Analytics	Pandas, NumPy, SciPy, Matplotlib, Data Cleaning, Time Series Analysis, Automated Excel / CSV Reports
AI Tools & Agents	Google Gemini API & Ecosystem, Antigravity IDE (Agentic AI pair-programming), Prompt Engineering for Code Automation

WORK EXPERIENCE

Independent Software Developer / Freelance Consultant

2018 – Present | Remote

Self-Employed / Freelance Technical Engagements

- **Enterprise Web Automation & RPA Pipeline (wbjit & Govt. Portals):** Architected and maintained Python + Selenium pipelines handling school-level fund allocations and automated portal submissions. Designed dynamic state tracking (processed_payees.txt) to guarantee fault-tolerant resumption of interrupted tasks without duplicate record submissions.
- **Cross-Platform Mobile App Development (Flutter & Dart):** Engineered responsive cross-platform applications with Flutter and Dart (flutter-project, multipage_test), implementing clean widget hierarchies, multi-page routing, and state handling.
- **Systems Programming & Performance Engineering (Rust):** Developed high-performance utilities and concurrency exercises in Rust (learning_rust), exploring memory-safe system routines and Cargo package tooling.
- **Algorithmic Trading & Financial Integration:** Engineered an automated algo-trading system utilizing the **ICICI Securities Breeze API**, streaming real-time market quotes, validating risk parameters, and auto-executing orders.
- **Full-Stack School Data Portal (PM-POSHAN Scheme):** Developed a complete multi-tier web application (Flask) for block-level school daily reporting. Built role-based access for Admins and Users, daily attendance tracking, automated calculations, and database backups.
- **Open-Source Python Tooling:** Developed and published FiancialYear on GitHub for fiscal accounting routines, along with desktop automation utilities.
- **AI-Assisted Workflow Engineering:** Integrated Google Gemini and Antigravity IDE into daily development, generating test suites and automating refactoring workflows.

Data Operations & Automation Specialist

2012 – 2022

Government & Educational Administrative Sectors

- Managed high-volume public sector databases, digital records, and institutional accounts across block and district administrations for 10+ years.
- Pioneered transition from manual, error-prone spreadsheets to automated Python data cleaning scripts and validation pipelines.
- Maintained 100% audit accuracy across large-scale state welfare schemes and financial allocations.

KEY GITHUB PROJECTS

Flutter Mobile Application Suite (flutter-project)

github.com/piidus/flutter-project →

Cross-platform mobile application developed with Flutter and Dart, demonstrating component-based layouts, navigation, and stateful widgets.

Systems Programming in Rust (learning_rust)

github.com/piidus/learning_rust →

Implementation of systems-level algorithms, ownership/borrowing patterns, concurrency, and CLI tooling using modern Rust.

Enterprise Automation Suite (wbjit / office_algo)

Python • Selenium • Threading

Automated government payee reconciliation and bulk allocation pipeline with automatic login, session retention, and dynamic resume mechanism.

Financial Year Package (FiancialYear)

github.com/piidus/FiancialYear →

Open-source Python library to compute Indian fiscal years, quarters, and accounting date intervals.

Data Science & Machine Learning Repository

github.com/piidus/data_science →

Jupyter notebooks, statistical models, predictive algorithms, and end-to-end data cleaning pipelines.

EDUCATION & CERTIFICATIONS

- **Bachelor of Arts (History Hons.)** – Netaji Subhas Open University (Passout Year: 2012)
- **Data Science Specialization Certificate** – PWSkills
- **100 Days of Code: Complete Python Pro Bootcamp** – Udemy
- **Google Apps Script & Web Automation** – Udemy
- **Diploma in Information Technology Application (DITA)** – Youth Computer Training Centre (Govt. of West Bengal)

CANDIDATE OVERVIEW FOR RECRUITER FAST-TRACK

Core Competencies	Python, Cloud (AWS), Flutter (Dart), Rust, Web Automation & Data Engineering
Total Experience	6+ Years in Software & Cloud Automation + 10 Years Data Operations
Work Mode Preference	Strictly 100% Remote / Freelance / Contract (Bedridden medical condition)
Notice Period	Immediate Joiner (0 Days)
AI Tools Expertise	Google Gemini API & Antigravity IDE (Agentic coding workflows)